WORKED SOLUTION

STEP II 2020, Question 12

Probability and Statistics • Biased Dice Inequalities

STEP II 2020, Question 12

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The probabilities must add up to 1, so the biases satisfy $\sum \varepsilon_i = 0$. Once that is written down, expanding $\sum(\frac{1}{n} + \varepsilon_i)^2$ makes every linear term vanish and leaves something that is visibly non-negative. Part (ii) is then part (i) in disguise: dividing positive numbers by their total turns them into the probabilities of some die.

Solution

Write $p_i = \mathrm{P}(X = i) = \frac{1}{n} + \varepsilon_i$. Since $\sum_{i=1}^{n} p_i = 1$,

$$1 = n \cdot \frac{1}{n} + \sum_{i=1}^{n} \varepsilon_i \quad \Longrightarrow \quad \sum_{i=1}^{n} \varepsilon_i = 0.$$

Also each $p_i \geq 0$, so $\varepsilon_i \geq -\frac{1}{n}$ for every i. All sums below run over $i = 1, \ldots, n$.

Part (i)

The two rolls are independent, so the probability that both show i is p_i^2, and these events are disjoint for different i. Hence

$$\begin{aligned} \mathrm{P}(\text{same score twice}) = \sum p_i^2 &= \sum \left(\frac{1}{n^2} + \frac{2\varepsilon_i}{n} + \varepsilon_i^2 \right) \\ &= \frac{1}{n} + \frac{2}{n} \sum \varepsilon_i + \sum \varepsilon_i^2 = \frac{1}{n} + \sum_{i=1}^{n} \varepsilon_i^2. \end{aligned}$$

For a fair die every $\varepsilon_i = 0$ and the probability is $\frac{1}{n}$. For a biased die not all the ε_i are zero, so $\sum \varepsilon_i^2 > 0$. A biased die is therefore *more* likely than a fair one to show the same score on two successive rolls.

Part (ii)

First rewrite the double sum. It adds $x_i x_j$ over all pairs with $j < i$, that is, over each unordered pair of distinct indices exactly once. Squaring $S = \sum x_i$ produces every x_i^2 once and every such pair twice, so

$$\sum_{i=2}^{n} \sum_{j=1}^{i-1} x_i x_j = \frac{1}{2}\left(S^2 - \sum x_i^2\right).$$

Ask yourself Where does a die come from, when the x_i are just any positive numbers?

Set $p_i = x_i / S$. These are positive and add up to 1, so they are the probabilities of some n-sided die. By part (i), $\sum p_i^2 = \frac{1}{n} + \sum \varepsilon_i^2 \geq \frac{1}{n}$ (with equality exactly when the die is fair, i.e. all x_i are equal). Multiplying by $S^2 > 0$,

$$\sum x_i^2 \geq \frac{S^2}{n}.$$

Therefore

$$\sum_{i=2}^{n} \sum_{j=1}^{i-1} x_i x_j = \frac{1}{2}\left(S^2 - \sum x_i^2\right) \leq \frac{1}{2}\left(S^2 - \frac{S^2}{n}\right) = \frac{n-1}{2n}\left(\sum_{i=1}^{n} x_i\right)^2,$$

which is the required inequality. Equality holds when all the x_i are equal.

Part (iii)

Now the probability is $\sum p_i^3$. Expanding, and using $\sum \varepsilon_i = 0$ again,

$$\begin{aligned}\sum p_i^3 &= \sum \left(\frac{1}{n^3} + \frac{3\varepsilon_i}{n^2} + \frac{3\varepsilon_i^2}{n} + \varepsilon_i^3\right) \\ &= \frac{1}{n^2} + \sum_{i=1}^{n}\left(\frac{3\varepsilon_i^2}{n} + \varepsilon_i^3\right) = \frac{1}{n^2} + \sum_{i=1}^{n} \varepsilon_i^2\left(\frac{3}{n} + \varepsilon_i\right).\end{aligned}$$

For a fair die this is $\frac{1}{n^2}$.

Ask yourself The cubes ε_i^3 can be negative. Could they outweigh the squares?

No, because the biases cannot be very negative: $p_i \geq 0$ gives $\varepsilon_i \geq -\frac{1}{n}$, so

$$\frac{3}{n} + \varepsilon_i \geq \frac{2}{n} > 0.$$

Each term $\varepsilon_i^2\left(\frac{3}{n} + \varepsilon_i\right)$ is therefore ≥ 0, and it is strictly positive whenever $\varepsilon_i \neq 0$, which happens for at least one i on a biased die. So

$$\sum p_i^3 > \frac{1}{n^2},$$

and a biased die is again *more* likely than a fair one to show the same score on three successive rolls.

Checks

- Take $n = 4$ with probabilities $\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$, so $\varepsilon = \frac{1}{4}, -\frac{1}{12}, -\frac{1}{12}, -\frac{1}{12}$. Directly, $\sum p_i^2 = \frac{1}{4} + \frac{3}{36} = \frac{1}{3}$, and $\frac{1}{4} + \sum \varepsilon_i^2 = \frac{1}{4} + \frac{1}{16} + \frac{3}{144} = \frac{1}{3}$, which beats the fair $\frac{1}{4}$. For three rolls, $\sum p_i^3 = \frac{1}{8} + \frac{3}{216} = \frac{5}{36} \approx 0.139$, well above the fair $\frac{1}{16} = 0.0625$.
- The most extreme die always shows 1: then both probabilities are 1, the largest possible.
- In (ii), with $n = 2$ the inequality reads $x_1 x_2 \leq \frac{1}{4}(x_1 + x_2)^2$, which is AM–GM. The inequalities in (i), (ii) and (iii) were also confirmed by computer on many thousands of random cases.

Common slips

- Forgetting that $\sum \varepsilon_i = 0$, and so being left with linear terms whose sign is unknown.
- In (iii), declaring $\sum \varepsilon_i^3$ to be "small" or of either sign without using $\varepsilon_i \geq -\frac{1}{n}$; that bound is what makes the argument work.
- In (ii), applying part (i) to the x_i themselves rather than to $x_i / \sum x_j$, which need not add up to 1.

Beyond STEP

The quantity $\sum p_i^2$ is the *collision probability* of a distribution, and part (i) says it is smallest for the uniform distribution; part (iii) is the same statement for $\sum p_i^3$. Both are cases of the convexity of $t \mapsto t^k$ (Jensen's inequality), or of the power-mean inequality, and the logarithm of $\sum p_i^k$ gives the Rényi entropies used in information theory and cryptography. Grimmett and Welsh, *Probability: An Introduction*, is a good place to meet these ideas in a probabilistic setting.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.